METR4202 crib sheet side 1 · estimation, mapping, models, perception, multi-robot

Exam: 20 MCQ (20) · SA (20) · 3 LA (40) · hurdle 40%. LA every year = Bayes update · planning by hand (A*/RRT*) · arm: M + screw axes + pinhole. SA every year = C-space labelling, Bayes 2 steps, sensor compare, implicit/explicit. MCQ = 'which is FALSE' → read ALL options.

Probability

- Product: $P(x,y) = P(x|y)P(y)$
- Independent: $P(x,y) = P(x)P(y)$. Cond indep: $P(x|y,z) = P(x|z)$ – once z known, y adds nothing.
- Total prob: $P(x) = \sum_y P(x|y)P(y)$; $\int$ for cont.
- Bayes: $P(x|z) = \eta \underbrace{P(z|x)}_{\text{likelihood}} \underbrace{P(x)}_{\text{prior}}$ / $\underbrace{P(x|z)}_{\text{posterior}}$; $\eta = 1/P(z)$: evidence
- Belief: $bel(x_t) = p(x_t|z_{1:t}, u_{1:t})$ = dist. over ALL states
- Markov: $p(x_t|x_{0:t-1}, z, u) = p(x_t|x_{t-1}, u_t)$; $p(z_t|x_{0:t}, \dots) = p(z_t|x_t)$. State is complete. Broken by: unmodelled history-dep. effects. (sensor heats up + noise grows, motor degrades) NOT broken by random noise (slip, gusts).

Bayes filter

$\overline{bel}(x_t) = \int p(x_t|u_t, x_{t-1}) bel(x_{t-1}) dx_{t-1}$
predict: action u uncertainty ↑
$bel(x_t) = \eta p(z_t|x_t) \overline{bel}(x_t)$
correct: sensor z, uncertainty ↓
Discrete: replace $\int$ by Σ over cells. Prediction never reduces uncertainty.

★ RECIPE A – discrete Bayes

1. List states with $P>0$
2. Predict: push each prob. through motion model. add when cells combine. Start colour irrelevant.
3. Correct: × | prior | $P(z|x)$ | product | ÷ Σ products = $P(z)$
4. $P(z|x)$ = entry for Observed z. $P(\text{white}|\text{black}) = 1 - P(\text{black}|\text{black})$. Classifier: use the column of detected class.
5. 2nd reading: posterior → new prior, repeat.
6. Decision: $E[\text{success}|a] = \Sigma_x P(s|x,a)P(x)$ → pick max. $P(s|\text{no object}) = 0$.
7. Check Σ=1. Comment on intuition (which reading was strong).

e.g. 2024: prior .4/.V.4/.1. $P(w|blk) = .17$, $P(w|wht) = .9$ → $P(z) = .535$ → .127/.032/.673/.168. 2025: after B .072/.904/.024, after A .483/.503/.013; suction .811 > pinch .743.

Kalman filter (linear + Gaussian)

$x_t = Ax_{t-1} + Bu_t + \varepsilon$, $\varepsilon \sim N(0,R)$ act. noise
$z_t = Cx_t + \delta$, $\delta \sim N(0,Q)$ → sensor noise
$\bar{\mu} = A\mu + Bu$, $\bar{\Sigma} = A\Sigma A^T + R$
$K = \bar{\Sigma}C^T(C\bar{\Sigma}C^T + Q)^{-1}$
$\mu = \bar{\mu} + K(z - C\bar{\mu})$, $\Sigma = (I - KC)\bar{\Sigma}$
1D: $K = \frac{\bar{\sigma}^2}{\bar{\sigma}^2+\sigma_z^2}$ Fuse 2 Gaus: $\mu = \frac{\sigma_2^2\mu_1 + \sigma_1^2\mu_2}{\sigma_1^2+\sigma_2^2}$
$\sigma^2 = \frac{\sigma_1^2\sigma_2^2}{\sigma_1^2+\sigma_2^2}$ (always smaller). K = trust dial:
Q↑ K↓ (ignore sensor) R↑ K↑ Q is inside K ⇒ changing sensor noise changes mean AND variance.
Optimal for linear-Gaussian, unimodal only.
EKF: nonlinear $g(u,x)$, $h(x)$: linearise with Jacobians $G = \partial g/\partial x$, $H = \partial h/\partial x$ at current μ: $\bar{\mu} = g(u,\mu)$
$K = \bar{\Sigma}H^T(H\bar{\Sigma}H^T + Q)^{-1}$, $\mu = \bar{\mu} + k(z - h(\bar{\mu}))$
Still unimodal: diverges if very nonlinear / bad init. guess / wrong data association.
Particle filter: belief = M weighted samples (any shape, multimodal). (1) sample $x_t^{(m)} \sim p(x_t|u_t, x_{t-1}^{(m)})$ (2) weight $w^{[m]} = p(z_t|x^{[m]})$ = sensor likelihood (3) resample ∝ w (low-variance sampler). Issues: cost ∝ M, grows exp. with dim; particle deprivation (add random particles); kidnapped robot.

	belief	use when
KF	1 Gaussian	linear, Gaussian, fast
EKF	1 Gaussian (linearised)	mildly nonlinear, good init (EKF-SLAM, GPS+odom)
PF	samples	multimodal, global loc., kidnapped, nonlinear, low-dim

Motion models (prediction step)

Velocity $u = (v, \omega)$ commanded → arc radius $r = v/\omega$:
$x' = x - \frac{v}{\omega}\sin\theta + \frac{v}{\omega}\sin(\theta + \omega\Delta t)$
$y' = y + \frac{v}{\omega}\cos\theta - \frac{v}{\omega}\cos(\theta + \omega\Delta t)$
$\theta' = \theta + \omega\Delta t + \gamma\Delta t$ (γ = final rot. noise)

★RECIPE – velocity model by hand
Compute $r = v/\omega$ (keep sign!) and $\omega\Delta t$ first · RADIANS · chain steps with new θ · ω=0: $x' = x + v\Delta t\cos\theta$, $y' = y + v\Delta t\sin\theta$. Check: (0,0,0), v=.8, ω=.03, 2s → (1.599, 0.048, 0.06). 2023: (1.1,−2.7,0) → (3.464,−2.343,0.3) → (4.453,−2.244,−0.1).

Odometry (encoders, proprioceptive): $\delta_{rot1} = atan2(\bar{y}' - \bar{y}, \bar{x}' - \bar{x}) - \bar{\theta}$, $\delta_{trans} = \sqrt{\Delta\bar{x}^2 + \Delta\bar{y}^2}$, $\delta_{rot2} = \bar{\theta}' - \bar{\theta} - \delta_{rot1}$; noise ∝ α params.

	velocity	odometry
data	command, before moving	measured, AFTER moving
planning?	YES (RRT steer fn)	NO
accuracy	lower	usually higher

Both used in prediction, not measurement. Constant (v,ω) ⇒ circular arcs.

Sensor models (correction step)

Beam: $p(z|x,m) = z_{hit}p_{hit} + z_{short}p_{short} + z_{max}p_{max} + z_{rand}p_{rand}$; beams indep. → product. hit = Gaussian about expected z^* (noise) · short = exponential $\lambda e^{-\lambda z}$, $z < z^*$ (unexpected objects) · max = spike at z_{max} (missed return: glass, black) · rand = uniform $1/z_{max}$. Weights Σ=1.
Likelihood field / scan: endpoint → distance to nearest obstacle, Gaussian; smooth, fast; ignores ray path.
Landmark: $r = \sqrt{\Delta x^2 + \Delta y^2}$, $\phi = atan2(\Delta y, \Delta x) - \theta$, + signature; compact but needs data association.
Characterise: many readings at known ranges → histogram → fit params (max likelihood / EM).

Mapping & SLAM

	pose	map	e.g.
Localisation	?	known	known tag map
Mapping	known	?	drone w/ perfect pose
SLAM	?	?	Moon rover, no GPS/map

Online SLAM $p(x_t, m|z_{1:t}, u_{1:t})$ · Full SLAM $p(x_{1:t}, m|\dots)$ (whole path).
Occupancy grid: static world, cells independent, binary. Prior 0.5 ↔ $l_0 = 0$ (uninformative). NEVER 0/1: certainty, can't ever update (0×anything, ±∞ log-odds).

$$p(m|z) = \frac{p(z|m)p(m)}{p(z|m)p(m) + p(z|\neg m)p(\neg m)}$$
$$l_t = l_{t-1} + \ln\frac{p(m|z_t)}{1-p(m|z_t)} - l_0, \quad p = 1 - \frac{1}{1+e^l}$$

★RECIPE B – occupancy cell
Table rows = true m / ¬m, cols = reading; rows sum to 1. FP = 1−TN, FN = 1−TP. Update reading by reading (posterior→prior). Log-odds step (prior .5) = $\ln\frac{p(z|m)}{p(z|\neg m)}$. Explain via likelihood ratio: occ reading LR = TP/FP, free reading LR = FN/TN.
TP .85, TN .99: occ → .9884, then free → .9287. TP .7, TN .95: .9333 → .8155 (occ LR 14 vs free 1/3.2 ⇒ stays high).

Data association: which landmark/place produced each measurement (or new?). Loop closure: recognise revisited place → constraint to old pose → drift spread back, map snaps consistent. Missed → drift grows, duplicated structure. False → map warps, overconfident Σ, may diverge, hard to undo (worst).
EKF-SLAM: joint Gaussian pose+N landmarks (3+2N), $O(N^2)$, single DA hypothesis. FastSLAM: particles for path + N small EKFs per particle (landmarks indep. given path), per-particle DA. Graph-SLAM: nodes = poses, edges = constraints, least-squares optimise; full SLAM; front-end (DA, loops) / back-end (optimiser).
Practical: dynamic objects, resolution vs memory, noise, stale maps, perceptual aliasing.

Perception

	measures	+	−
camera (passive)	dense colour, bearing only	cheap, semantics	no depth, lighting
lidar (active ToF)	range+bearing	metric, dark OK	sparse, no colour, costly; 2D = 1 plane
sonar	range	very cheap	wide beam, specular
IMU/encoder (proprio)	accel, rate / wheel rot	high rate	drift

Camera = WHAT things are; lidar = WHERE surfaces are. Lidar uses: occupancy maps, SLAM, obstacles. Camera: recognition, fiducials, lanes, servoing.

★RECIPE G – pinhole calc
$h' = \frac{f}{Z}h \Rightarrow Z = \frac{fh}{h'}$, $f = \frac{h'Z}{h}$
Camera pos = p of M; optical axis = $\hat{z}_b$. Z = depth ALONG optical axis from camera (not from base; add/subtract link offsets). Convert to metres. Assume: ideal pinhole, no distortion, object centred & ⊥ axis, flat. Size ∝ 1/Z (sanity ratio check).
JWST: 131.4×2.82e8/1.2e12 = 0.031 m. 2024: Z=05·60/.038=78.95 m (+1.5 to base); 20 m closer → 50.9 mm. 2025: $\sqrt{2.7^2 - 2^2} = 1.85$ m → f=80 mm.

$\lambda[u, v, 1]^T = K[R|t]X_w$, $K = \begin{bmatrix} a_u & 0 & u_0 \\ 0 & a_v & v_0 \\ 0 & 0 & 1 \end{bmatrix}$,
$u = a_u X/Z + u_0$. Intrinsic: f, pixel size, principal pt, skew, distortion. Extrinsic: camera 3D position + orientation (R,t).
Perspective keeps lines, conics; NOT angles, lengths, parallelism (vanishing pts); far = small; scale ambiguity (one image: relative size ✓, absolute size/depth ✗). Stereo $Z = fb/d$.
Calibration: checkerboard many poses → corners → solve K, distortion, extrinsics; min. reprojection error.
PBVS: estimate target 3D pose, Cartesian error; needs accurate calib incl. known cam→EE transform; straight 3D path, target may leave FOV. IBVS: error on image features $e = s - s^*$, $v = -\lambda L^+ e$; robust to calib (cam-EE just fixed), keeps features in view, odd 3D path, needs depth est.
Fiducial: 1 marker enough in theory (4 corners → PnP); more = average noise, resolve ambiguity, occlusion robust.

Multi-robot

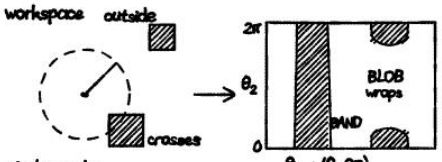
Harder: task allocation, duplication, collisions/congestion, comms limits, scalability, unknown teammate intent, credit assignment (each robot's contribution to team reward).
Implicit (infer from observation, no msgs): + robust to comms loss, cheap, scales; − wrong inference ⇒ duplication/conflict, less efficient.
Explicit (communicate/negotiate, auctions): + avoids conflicts, near-optimal allocation, shared maps; − bandwidth, latency, msgs grow with N, fails if network drops.
Learning-based (MARL, centralised training / decentralised exec., joint state-action value, credit assignment e.g. difference rewards) vs belief-space planning (Dec-POMDP: beliefs about world + teammates; principled, handles uncertainty, very expensive as N grows).

METR4202 crib sheet side 2 · planning, kinematics, RL, MCQ traps

Planning basics

Workspace = physical space. C-space = joint/config values; robot = point; C-obstacles; C_{free}. Straight free line in workspace ≠ robot can move (body/links sweep volume) → check in C-space. Hierarchy: global (A*, PRM; slow, rare) + local/reactive (DWA, VO, fields; fast, disturbances). Grid = resolution complete; PRM/RRT = probabilistically complete; PRM*/RRT* = asymptotically optimal.

★RECIPE C – C-space labelling (2-link arm) SA 3–5 mk



1. Draw circle radius L_1 (link-1 reach) about base.
2. Obstacle CROSSES circle ⇒ link 1 hits for any θ_2 ⇒ full-height vertical BAND over θ_1 pointing at it (+ bulges = link 2).
3. Obstacle OUTSIDE ⇒ link 2 only ⇒ BLOB near $\theta_2 \approx 0 \equiv 2\pi$ (arm straight), $\theta_1 \approx$ obstacle's bearing from base; wraps top↔bottom.
4. Bearing of obstacle edges from base = atan2 → θ_1 range.
5. Check a point: elbow $= b + L_1(c_1, s_1)$, tip = elbow $+ L_2(c_{12}, s_{12})$ inside obstacle? Both axes wrap $(0 \equiv 2\pi)$.

Graph search

	expands	optimal?
BFS	FIFO oldest	fewest edges only; = Dijkstra if uniform cost
DFS	LIFO newest	no (low memory)
Dijkstra	min g (cost-to-arrive)	yes (costs ≥0); = A* with $h=0$
Greedy	min h	no
A*	min $f = g + h$	yes if h admissible

Admissible: $h(n) \le h^*(n)$ never overestimates. Consistent: $h(n) \le c(n, n') + h(n')$. Straight line ✓ for roads/Euclidean. Manhattan ✗ if diagonal/free motion: (0,0)→(1,1): $2 > \sqrt{2}$. Inflated h (eh) faster, suboptimal.

★RECIPE D – A* by hand 10 mk

Tables each step: OPEN: node | f=g+h | parent and CLOSED: node | g | parent.

1. Pop lowest f → CLOSED.
2. Each neighbour: $g = g_{cur} + c$. New → add. In OPEN with higher g → replace g + parent. In CLOSED → skip.
3. STOP when goal is POPPED (not when first added).
4. Backtrack parents; state path + cost; quick sanity vs next-best route.

2024: KG→GP→SB→MG = 3.1+1.0+9.7 = 13.8 km (SL parent changed H→SB).

Sampling-based

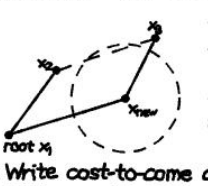
Avoid explicit C-space (high-DOF). Need: sampler, distance metric, local planner, collision checker (edges checked along WHOLE length, incl. footprint – not just endpoints).

PRM (multi-query): sample free q's → connect k-nearest/radius r with collision-free local paths → roadmap → graph search per query.

RRT (single-query): sample q_{rand} (goal bias) → nearest node → steer step → collision check → add. Not optimal (early jagged branches locked in).

RRT*: choose-parent + rewire within radius $r \Rightarrow$ asymptotically optimal.

★RECIPE E – RRT vs RRT* 10 mk



1. Write cost-to-come $c(x)$ for every node FIRST.
2. RRT parent = nearest (distance only).
3. RRT* parent = $\arg\min_{near} c(x_{near}) + d(x_{near}, x_{new})$ (collision-free).
4. Rewire each other near node if $c(x_{new}) + d(x_{new}, x_{near}) < c(x_{near})$: delete old parent edge, add new; descendants' costs drop too.

2025: parent x_6 (2.8+2.2=5.0); rewire x_4 9.9→8.5.

Local: Potential field $U = \frac{1}{2}\zeta d_{goal}^2 + \frac{1}{2}\eta\left(\frac{1}{D} - \frac{1}{Q^*}\right)^2$ ($D \le Q^*$), $\dot{q} = -\nabla U$; local minima (U-shaped traps).

DWA: sample (v, ω) in reachable window $[v \pm a\Delta t]$, admissible (can stop), simulate short arcs, maximise α heading + β clearance + γ velocity.

VO: velocities that collide with moving obstacle within horizon τ; pick v outside VO nearest preferred.

Design Q (e.g. warehouse holonomic fleet): global planner on known map (A* on grid/roadmap, PRM reused = multi-query) + local robot-robot avoidance (VO/DWA) + justify.

MDP = planning under uncertainty (stochastic outcomes) → policy.

Kinematics

FK: joints → EE pose. IK: pose → joints. (Robot pose after velocity cmds = FK; reach door handle = IK.) Grübler: dof $= m(N-1-J) + \sum f_i$ (m=3 planar, 6 spatial; N incl. ground; pin joining k links = k-1 joints). Rigid body: 3 DOF planar, 6 spatial; R,P joints 1 DOF.

$$R_z = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix} \; R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix} \; R_y = \begin{bmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{bmatrix}$$

$T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}$, $T^{-1} = \begin{bmatrix} R^T & -R^T p \\ 0 & 1 \end{bmatrix}$. $[Ad_T] = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix}$. $SO(3)$ rot, $SE(3)$ transf., $so(3)$ skew $[\omega]$. $se(3)$ twist mats.

space: $T = e^{[S_1]\theta_1} \cdots e^{[S_n]\theta_n} M$ · body: $T = M e^{[B_1]\theta_1} \cdots e^{[B_n]\theta_n}$

Rodrigues $e^{[\omega]\theta} = I + \sin\theta[\omega] + (1 - \cos\theta)[\omega]^2$. $e^{[S]\theta}$: rot part as above, trans $(I\theta + (1-c)[\omega] + (\theta - s)[\omega]^2)v$; prismatic: $[I, v\theta]$. $B_i = (Ad_{M^{-1}})S_i$.

★RECIPE F – arm LA 15 mk: M, screw table, camera

1. Pin {s} axes: which arrow is OUT of page (drawn diagonal). Right-handed: $\hat{x} \times \hat{y} = \hat{z}$.
2. M: cols of $R = \hat{x}_b, \hat{y}_b, \hat{z}_b$ in {s} coords; p = {b} origin (sum link lengths along axes). Check $\det R = +1$.
3. Revolute: ω = unit axis (right-hand rule w/ drawn arrow), q = ANY point on axis, $v = -\omega \times q$. Prismatic: $\omega = 0$, v = unit travel dir.
4. {b}-frame: ω, q in {b} coords measured from {b} origin, same formula.
5. Table i | ω | q | v. Flip $\omega \Rightarrow$ flip v (state your arrow reading).
6. Camera part → RECIPE G (side 1) with p, $\hat{z}_b$ from M.

ω	$v = -\omega \times q = q \times \omega$
(1,0,0)	$(0, q_z, -q_y)$
(0,1,0)	$(-q_z, 0, q_x)$
(0,0,1)	$(q_y, -q_x, 0)$

Check: $\omega = (1,0,0)$, $q = (0, 0, L_1) \rightarrow v = (0, L_1, 0)$. Cross: $a \times b = (a_y b_z - a_z b_y, a_z b_x - a_x b_z, a_x b_y - a_y b_x)$.

Jacobian $V = J(\theta)\dot{\theta}$; $J_{s1} = S_1$, $J_{s2} = [Ad_{e^{[S_1]\theta_1}}]S_2$; $J_b = [Ad_{T_{ba}}]J_s$. Singularity: J loses rank → can't move in some direction (frame-independent). Manipulability ellipsoid from JJ^T (velocity); force ellipsoid ⊥.

Closed-form IK (Pieper): 6-DOF, 3 consecutive revolute axes intersect at a point (spherical wrist) or 3 consecutive parallel ⇒ position/orientation decouple.

Newton-Raphson IK: guess θ_0; $[V_b] = \log(T_{sb}^{-1}(\theta_i)T_{sd})$: $\theta_{i+1} = \theta_i + J_b V_b$; until $\|\omega_b\|, \|v_b\| < \varepsilon$. Needs good initial guess; may find other solution; trouble near singularities.

$J^\dagger = J^T(JJ^T)^{-1}$ (fat).

IK solutions: 0 (outside workspace) · finite (non-redundant) · ∞ (redundant).

Reinforcement learning

MDP (S, A, T, R, γ) – NOT timestep, NOT return. Policy $\pi : S \to A$. Goal: maximise expected return $G_t = \sum_k \gamma^k r_{t+k+1}$ (sum. discounted – not immediate/final/avg).

$V^\pi(s) = E[G|s]$. $Q^\pi(s,a) = E[G|s,a]$, $\pi^*(s) = \arg\max_a Q^*(s,a)$.

$Q^*(s,a) = \sum_{s'} T(s,a,s')(R + \gamma \max_{a'} Q^*(s',a'))$

Value iteration: repeat Bellman backup (needs model T). TD(0): $V(s) \leftarrow V(s) + \alpha[r + \gamma V(s') - V(s)]$ (model-free).

Q-learning (off-policy): $Q \leftarrow Q + \alpha[r + \gamma \max_{a'} Q(s',a') - Q]$; SARSA (on-policy) uses actual a'. ε-greedy: explore vs exploit.

$\gamma \in [0,1)$: small = myopic, →1 far-sighted; keeps sums finite. α = learning rate.

Markov ⇒ RL state must hold all info to predict next (pos + vel, battery…) else partially observable.

RL vs planner: policy = action for EVERY state, closed-loop, learned from reward, any start; planner = 1 open-loop path from model; RL costs training, no guarantees.

Challenges: sample inefficiency, unsafe exploration, reward design, partial obs., continuous spaces, sim-to-real gap (friction, contact, latency, noise mis-modelled). Fixes: domain randomisation, system ID, fine-tune on hardware, demonstrations/imitation, reward shaping.

Sim: + safe, fast, parallel, free resets, perfect state; – reality gap.

MCQ traps (all from past papers)

- Odometry model: proprioceptive, prediction step, more accurate, CAN'T plan. Velocity model: CAN plan, prediction step.
- RRT = single-query; PRM multi; sampling needs no explicit C-space but needs collision check.
- BFS = Dijkstra for uniform costs; A* ≠ BFS. Dijkstra expands by cost-to-arrive.
- PF weight = sensor likelihood. KF belief = Gaussian. Sensor noise $\Delta \Rightarrow$ mean+var change.
- Return/timestep ≠ MDP. Policy: states→actions; max cumulative discounted reward: $\pi^* = \arg\max_a Q^*$.
- Focal length = intrinsic; position/orientation = extrinsic. Angles not preserved; parallel lines meet.
- Sim-to-real = sim mis-models contact/friction/slip. Credit assignment = each robot's contribution.
- Explicit = communicate/negotiate; implicit = infer from observation.
- Global/local hierarchy: global rarely (expensive), local fast for disturbances ⇒ "both".